

| National University of Computer and Emerging Sciences, Lahore Campus              |          |                               |              |             |
|-----------------------------------------------------------------------------------|----------|-------------------------------|--------------|-------------|
|  | Course:  | Machine Learning for Robotics | Course Code: | CS 5096     |
|                                                                                   | Program: | BS (Computer Science)         | Semester:    | Spring 2026 |
|                                                                                   | Date:    | 27-04-2026                    | Total Marks: | 10          |
|                                                                                   | Section: | A                             | Weight:      |             |
|                                                                                   | Exam:    | Quiz 2                        | Roll No:     |             |

Q1) Consider a neural network using a unit step activation function. (6)

$$f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

Find a valid combination of weights ( $w_1, w_2$ ) and bias ( $b$ ) for each of the following gates. For each gate, show the calculation for the input pair (1,0) to verify your logic. Constraints: The absolute value of  $w_1$  and  $w_2$  must be prime.

a) The OR Gate

To satisfy the OR logic, the weighted sum ( $z$ ) must cross the 0 threshold for any input that isn't (0,0).

- (0,0)→0:  $f(b)=0 \rightarrow b \leq 0$
- (1,0)→1:  $f(w_1+b)=1 \rightarrow w_1+b > 0$
- (0,1)→1:  $f(w_2+b)=1 \rightarrow w_2+b > 0$
- (1,1)→1:  $f(w_1+w_2+b)=1 \rightarrow w_1+w_2+b > 0$
- $|w_1|$  and  $|w_2|$  must be prime

Example solution:

$$w_1 = 3, w_2 = 3, b = -1.5$$

1.5 marks for correct ( $w_1, w_2, b$ )

$$3(1) + 3(0) - 1.5 = 1.5$$

1 mark for correct calculation of  $w_1 \cdot x_1 + w_2 \cdot x_2 + b$

$$f(1.5) = 1$$

0.5 marks for  $f(z)$

b) The NAND Gate (Output is 0 only if both inputs are 1)

Must satisfy:

- $(0,0) \rightarrow 1: f(b)=1 \rightarrow b > 0$
- $(1,0) \rightarrow 1: f(w_1+b)=1 \rightarrow w_1+b > 0$
- $(0,1) \rightarrow 1: f(w_2+b)=1 \rightarrow w_2+b > 0$
- $(1,1) \rightarrow 0: f(w_1+w_2+b)=0 \rightarrow w_1+w_2+b \leq 0$
- $|w_1|$  and  $|w_2|$  must be prime

Example solution:

$$w_1 = -3, w_2 = -3, b = 5$$

$$-3(1) - 3(0) + 5 = 2$$

$$f(2) = 1$$

Q2) You are designing a neural network for a computer vision task. The network processes a batch of images and passes them through two hidden layers. The input  $X$  is a batch of 50 images, where each image is flattened into a vector of 1,024 pixels.

Layer 1 (L1) has 256 neurons.

Layer 2 (L2) has 128 neurons.

Output Layer (L3) is a multiclass classification layer with 10 units (one for each class).

The network is implemented using the vectorized formula:

$$Z = W \cdot X + b$$

What are the dimensions of the input matrix  $X$ ?

(2)

1024 x 50

What are the dimensions of the weight matrix ( $W[1]$ ) and bias vector ( $b[1]$ ) for the first layer?

(2)

Weight matrix dimensions : 256 x 1024

Bias vector dimensions: 256 x 1